77c88721-5cb7-4690-a34c-a85b7eb0a74f

<prev_user>

<request>

You are working on an issue in the 'martincostello/alexa-london-travel-site' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/alexa-london-travel-site/alexa-london-travel-site (not in /tmp/inputs).
Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-03T12:32:45.047Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Investigate why the UI tests using Playwright in the build GitHub Actions workflow keep failing from flakiness on Ubuntu and propose relevant fixes to improve the test stability.
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.

<prev_assistant>

Summary

Here's what I found and fixed:

Root Cause
All failures on Ubuntu Linux showed `TargetClosedException: Target page, context or browser has been closed` from Playwright. This is caused by **Chromium renderer process crashes due to limited `/dev/shm` (shared memory)** on GitHub Actions Ubuntu runners.

Evidence from artifact analysis:
- Every failure on Linux is a `TargetClosedException` — different tests fail randomly each run (twitter, apple, google, chromium null channel, chrome channel)
- Firefox tests **never** fail — Firefox doesn't use `/dev/shm` the same way Chromium nt on those platforms
- Multiple workflow runs required 6–12 re-attempts before passing (`run_number: 8099` needed **8 attempts**)
- Trace analysis confirmed that 3+ seconds of server-side processing during Alexa token deletion coincides with browser crashes

Changes Made

`BrowserFixture.cs` — Added `--disable-dev-shm-usage` to Chromium launch options on Linux:
```csharp
else if (OperatingSystem.IsLinux() && Options.BrowserType == BrowserType.Chromium)
{
    // ...
    options.Args = ["--disable-dev-shm-usage"];
}
```
This tells Chrome to use `/tmp` instead of `/dev/shm`, bypassing the shared memory size limitation. Scoped to Linux + Chromium only (Firefox and other platforms unaffected).

`PreferencesTests.cs` — Replaced two `Task.Delay(TimeSpan.FromSeconds(1), TestContext.Current.CancellationToken)` calls with `await page.WaitForLinesAsync()`. Trace analysis confirmed the line preference elements appear ~80ms after the sign-out button, so a Playwright selector wait is both more reliable and faster than a fixed 1-second delay.

Security Summary
No security vulnerabilities were introduced or found. The `--disable-dev-shm-usage` flag has no security implications — it only changes which filesystem location Chrome uses for shared memory. (The initially

considered `--no-sandbox` flag was not included as it would unnecessarily weaken the browser's security sandbox.)

<current>

<request>

You are working on an issue in the 'martincostello/alexa-london-travel-site' repository.

<repository_context>
I've cloned the repository in the directory /home/runner/work/alexa-london-travel-site/alexa-london-travel-site (not in /tmp/inputs). Always use absolute paths when referring to files in the repository.
</repository_context>

<current_datetime>2026-03-03T13:58:01.078Z</current_datetime>

Consider the following problem statement:

<problem_statement>
Looks like there's still failures: https://github.com/martincostello/alexa-london-travel-site/actions/runs/22626117483/job/65563117104?pr=4057
</problem_statement>

Implement the necessary changes to the repository so that the requirements specified in the <problem_statement> are met.

A **CI/Build Failure** has been mentioned in the problem statement, make sure to follow your instructions for addressing CI/Build Failures.